David Culver

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EXPERIENCE

Undergraduate Research Assistant — PI: Matthias Morzfeld **March 2024 - Present**
Scripps Institution of Oceanography, La Jolla, CA

- Focused on constructing a lightweight (5×10^5 parameters) coordinate based Neural Network for implicit neural representations (or neural fields) to work under computational constraints imposed under hardware limits (working on NVIDIA GTX 1650 GPUs w/ 4Gb of vram each).
- Added derivative-based regularization to the training objective, establishing whether a physics-informed neural network (PINN) formulation improved accuracy under the model's size constraints.
- Moved training to GPU using PyTorch/CUDA for an 5x throughput gain, submitting long-running jobs to the SIO compute cluster via SLURM.
- Gave successful presentations to groups of various levels of scientific machine learning knowledge, from nontechnical groups to professional groups.

Undergraduate Research Assistant — PI: Falk Feddersen **January 2025- Present**
Scripps Institution of Oceanography, La Jolla, CA

- Extended the coordinate-based neural field framework to nearshore breaking waves, fitting LiDAR scans collected by hover drone at a remote field site.
- Modeled up to 60 s of LiDAR data ($\approx 10^7$ points) to within the 6 cm sensor tolerance, producing a continuous, differentiable sea surface that can be evaluated at arbitrary space–time coordinates rather than only where sampled.
- Recovered summary wave statistics, including wave speed and wave height, directly from the learned model in agreement with established analytical methods.
- Authored a first-author paper in collaboration with Matthias Morzfeld and Falk Fedderson, and presented the results to an audience of professional oceanographers.

Reader and Instructional Assistant —Class: Introduction to Ocean Waves **March 2026 - June 2026**
UC San Diego, CA

- Consistently graded work on time and hosted weekly office hours.
- Communicated with the professor of the class what students needed to succeed.
- Was friendly and inclusive to all students who needed assistance

UCSD Club Treasurer — Men's Club Water Polo **September 2023 - June 2026**
UC San Diego, CA

- Prepared and delivered budget reports based on club spending to the administration at UC San Diego
- Managed all finance in custom Excel documents, and made clear and decisive purchase's for the club.
- Helped manage and lead all members of the water polo club by focusing on making the club a place for anyone who cared to join.
- Focused on maintaining a level of excellence within the club.

Random Matrix Representation Theory **March 2023-June 2023**
San Diego, CA

- Participated in an advanced reading group focused on rigorous approaches to Principal Component Analysis (PCA) and its applications to random matrix theory.

Forestry Technician — Coastal Forestry **June 2024 - September 2024**
Santa Cruz, CA

- Worked as a field technician for general forestry work including basic surveying, tree painting, and mapping in GIS.
- Maintained a consistent schedule and made it in on time.

Grocery Stocker — Shoppers Corner **March 2022 - September 2022**
Santa Cruz, CA

- Worked in the dairy section of a locally owned grocery store.
- Stocked all dairy products and alcoholic beverages by making hourly rounds and removed all items which had expired.
- Consistently was on time and maintained a reliable working schedule.

TALKS AND PRESENTATIONS

Differentiable Sea Surface Representations from Neural Network Modeling LiDAR Point Clouds May 2026
Undergraduate Research Conference (URC), UC San Diego — Contributed talk

Using Neural Networks as Functional Representation of the Sea Surface from a set of irregular LiDAR October 2025
ROSXI (Rocky shOrlines eXperiment SImulations), La Jolla, CA — Contributed talk

Functional Representation of the Sea Surface using Neural Networks September 2025
S^3 (Scripps Student Symposium), La Jolla, CA — Contributed talk

Using neural networks for functional representations of nearshore ocean waves July 2025
CaCAO (Chaos, Computation, Analysis, and, Optimization), La Jolla, CA — Contributed talk

Implicit Neural Representations of Nearshore Ocean Waves March 2025, March 2026
SoCAMS (Southern California Applied Mathematics Symposium) — Poster

PROJECTS

Coordinate-Based Neural Field Ocean Wave Model — PyTorch, CUDA, SLURM **2024 - 2026**
Independent research project, Scripps Institution of Oceanography

- Built a 5×10^5-parameter coordinate-based network that maps space–time coordinates directly to sea-surface elevation, trainable on a single 4 GB GPU.
- Fit $\approx 10^7$ drone-LiDAR points to within the 6 cm sensor tolerance, yielding a continuous, differentiable surface evaluable off the sample grid.
- Extracted wave speed and wave height analytically from the learned field, matching established methods.

Skin Lesion Classifier (CNN) — TensorFlow, Python, Git **PHYS 139, 2025**
Team project, UC San Diego

- Trained a lightweight ResNet-based CNN to classify skin tumors as malignant or benign, fine-tuned with dropout regularization to control overfitting on a small dataset.

- Validated with cross-validation and standard classification metrics (ROC–AUC, precision/recall) rather than accuracy alone.
- Ran training on a cloud VM and coordinated a multi-person codebase through Git/GitHub.

Automatic Waffle Maker — C++, Arduino, SolidWorks **2024**
Team project, UC San Diego

- Designed the full mechanism in SolidWorks and fabricated custom 3D-printed parts for batter dispensing and iron actuation.
- Wrote the C++ firmware driving stepper motors and servos through a timed cook cycle on Arduino microcontrollers.
- Collaborated on hardware and firmware with version control via Git/GitHub.

EDUCATION

University of California: San Diego — **GPA:** 3.877 Graduated June 2026
Physics, BS **Major GPA**: 3.879
Mathematics, BS **Major GPA**: 3.975